

Scarcity Is Not a Stock

AI memory and power: what today's prices already assume

Micron Technology (MU) · Vistra Corp. (VST) · PJM capacity market. Market data as of the 15 September 2026 close.

INVESTMENT VIEW

Micron: no demonstrated valuation edge

At \$927.60 the shares require a steady-state DRAM gross margin of about 70% once the shortage has passed. Micron's best year before this cycle was 58.9%, and its best quarter 61.0%. Our FY2027 estimate is above consensus, so the disagreement is not about next year. It is about whether a capital-heavy commodity business can hold a margin nine to eleven points above its record after new fabs arrive. The public evidence does not establish that. We are not calling the shares overvalued: the price is a fair bet if the shortage is as deep and as long as the most bullish insiders say.

PJM and Vistra: no capacity-driven edge

PJM is short of capacity, but a regulatory collar sets the price through 2029/30 and a backstop procurement is contracting new supply. The open question, the price after the collar, moves Vistra's equity by about 1% for each year it lasts, and by -6% to +9% only if permanent.

THE DEBATE IN ONE NUMBER

How short is the industry today? TrendForce puts 2026 DRAM sufficiency at -1% to -2%; on that evidence the shortage closes around 2030 and the price needs ~70%. SK Group's chairman says wafer supply trails demand by more than 20% until about 2030; on that view the requirement falls to the mid-50s to low-60s, near the record. The price is betting on the second.

VARIANT PERCEPTION

Consensus	Earnings stay exceptionally strong through FY2027 (EPS \$155; 6.0× forward).
Our near-term view	Achievable. Base case about \$168. Near-term earnings are the same at any steady-state margin, so the valuation turns on what follows.
Where we differ	At \$927.60 the stock trades at 19× the FY2030–33 earnings the price itself requires.
What would prove us wrong	Fabs late at two or more suppliers while demand holds; HBM above 35% of wafer starts in 2028; contract renewals at high floors.
What would prove us right	Contract prices peaking in 1H CY2027 with Yongin, ID1 and Tongluo on schedule; FY2030 gross margin below 60%.

MICRON (MU)

Share price, 15 Sep 2026	\$927.60
Equity value (1.145bn diluted)	~\$1.06tn
FY2027 EPS, consensus	\$155
FY2027 NOPAT/share, base case	~\$168
Required steady-state DRAM GM	~70%
Record year / quarter (FY2018)	58.9% / 61.0%
Value at the record margin	\$682
Value at FY17–25 average (35%)	\$416
P/E on FY2030–33 required NOPAT	19×

PJM CAPACITY AND VISTRA (VST)

VST share price, 15 Sep 2026	\$141.53
2028/29 clearing price (= cap)	\$325.00
Shortfall vs requirement	6.8 GW
Vistra cleared, 2028/29	10.9 GW
Merchant after Meta PPAs	8.8 GW
Equity effect, per year of outcome	-0.6% to +0.9%

NEXT DATA POINTS

Micron FY2026 results, FY2027 outlook	30 Sep 2026
PJM backstop procurement opens	30 Sep 2026
Hyperscaler capex guidance	Oct–Nov 2026
DRAM contract prices; ID1, Yongin, Tongluo	1H CY2027

Author: [name, title, registration, contact]

Exhibit 1: Value and implied return by steady-state DRAM margin

Steady-state DRAM gross margin	Value/share	vs price	Implied return
35% (FY2017–25 average)	\$416	–55%	5.0%
50%	\$555	–40%	7.1%
58.9% (pre-2026 record year)	\$682	–26%	8.6%
69.9% (required by the price)	\$928	0%	11.0%
75%	\$1,105	+19%	12.4%
80%	\$1,347	+45%	14.0%

Source: report model. Cost of equity 11%. Every row includes the same FY2027–29 shortage cash. Implied return is the discount rate at which value equals \$927.60. At historical-average margins a buyer earns about the 10-year Treasury yield (5.00%), with no equity risk premium.

Exhibit 2: Without the industry model: FY2030 operating margin the price needs, by FY2030 revenue

FY2030 revenue	Growth 3% after	Growth 5% after
\$100bn	99%	76%
\$125bn	79%	61%
\$150bn	66%	51%
\$200bn	49%	38%
\$250bn	39%	31%
Required operating profit	\$99bn	\$76bn

Operating value \$1,032bn less the present value of FY2027–29 free cash flow (\$302bn) is what FY2030 onward must earn. Return on new capital 64% (the model's steady state; at 30%, \$85–104bn), tax 15%, 11% cost of equity. FY2018: \$30.4bn revenue, 49% operating margin; FY2025: \$37.4bn, 26%. Class D.

Exhibit 2 reduces the question to two numbers investors can observe: Micron's revenue once prices normalise, and the operating margin it keeps on it. The price works if Micron is roughly a \$150bn business at a 50–65% operating margin, or a \$200bn business at 40–50%. Its largest pre-2026 year was \$37bn.

The method is not circular. The model's FY2027 NOPAT is \$192bn whether the steady-state margin is 59%, 70% or 80%: near-term earnings come from the starting point and the shortage path, not from the solved margin. The solved margin governs only what happens after the shortage. We then judge it against history and economics, not against the model that produced it.

1. Framework

The title is the thesis: a shortage becomes an investment only when the price has not already paid for it. A bottleneck is a fact about an industry; a share price is a claim about future cash. Once the market can see a constraint, an investor is paid only for what the price has not yet assumed. We start from the price, solve for the operating state it implies, and test that state against filings, contracts, competitors' plans and the physics of the supply chain.

Exhibit 3: Seven questions a constraint thesis must answer, in order

#	Question	Memory (Micron)	Power (Vistra)
1	What exactly is scarce?	DRAM wafers, drawn down by HBM	Accredited capacity (UCAP)
2	What would the owner earn without it?	FY2017–25 average margin, 35%	Pre-2024 price, \$28.92/MW-day
3	Who keeps the difference?	Suppliers, until new fabs land	New entrants via 15-year contracts; incumbents only after the collar
4	How fast does it erode?	Announced fabs close the gap around 2030	Collar to 2029/30; backstop adds supply
5	What capital keeps it?	Heavy: capex ~52% of cost of goods	Low for existing plants
6	What does the price assume?	~70% margin after the shortage	Capacity revenue at the capped level, known to 2028/29
7	What shows the answer first?	Contract prices, fab milestones, hyperscaler capex	Backstop results; 2030/31 market design

Evidence classes used in every exhibit: F = fact reported by a company, market operator or regulator; E = external estimate (industry tracker, operator estimate, consensus); D = derived from F and E; A = analyst assumption.

2. Memory

2.1 What happened: a price-and-mix event

In the third quarter of FY2026, DRAM revenue of \$31.3bn rose 67% from the prior quarter on bit shipments up in the low single digits and prices up in the low 60s per cent. NAND prices rose in the mid 80s per cent on bits up in the mid single digits.

Almost all of the gain was price. Disclosure does not split price from mix (HBM, server) or utilisation; how long the price-and-

mix gain lasts is the valuation question.

Exhibit 4: Revenue rose roughly eightfold in three years, almost entirely through price

	FY2018	FY2023	FY2025	FY2026E	Q4 FY2026 guide, annualised	Type
Revenue	\$30.4bn	\$15.5bn	\$37.4bn	~\$129bn	~\$200bn	F / D
Gross margin (GAAP)	58.9%	-9.1%	39.8%	~80%	~86%	F / D
Operating income	\$15.0bn		\$9.8bn			F
Capex, net of incentives			\$13.8bn	~\$27bn		F

Source: Micron FY2018, FY2025 and Q3 FY2026 releases; Q3 FY2026 prepared remarks. FY2018 Q4 GAAP gross margin 61.0%. FY2026E = nine months reported (\$79.0bn) plus Q4 guidance of \$50.0bn ± \$1.0bn. Q4 guidance also: ~\$1.65bn operating expenses, ~15% tax rate, ~1.15bn shares.

2.2 Why supply responds slowly

HBM, the memory stacked beside AI accelerators, uses the same wafers as server and phone DRAM. TrendForce puts HBM at 22% of the top three suppliers' DRAM wafer input and 9% of their bit supply at the end of 2026, and 30% and 13% at the end of 2027. The wafer used per bit, relative to the rest of DRAM, is therefore $(22/78) \div (9/91) = 2.85$ at end-2026 and $(30/70) \div (13/87) = 2.87$ at end-2027. The comparison is with a mixed basket of DDR5, DDR4 and LPDDR, not a single product. The model holds 2.87 through 2029, 2.85 in 2030 and 2.8 from 2031, because no source supports a further rise.

HBM also needs advanced packaging and base-die logic capacity. If packaging binds first, HBM output is capped regardless of wafers: HBM prices stay firm, less wafer capacity is diverted, and the conventional shortage weakens. Our model prices HBM at wafer parity and captures only the conventional channel, so it neither credits Micron with an HBM premium nor charges it for a packaging cap.

2.3 Supply: what is disclosed, and what we assume

Exhibit 5: Disclosed or externally estimated DRAM capacity plans

Supplier	Announced additions	Timing	Type	Source
SK hynix	~550k → ~1m wafers/month; M15X; Yongin	2030–31	E	TrendForce, 5 Jun 2026 (reported; not company-confirmed)
Micron	Tongluo phase 1 (>10% of Q4 2026 capacity); Idaho ID1, ID2; New York	2H27; mid-27; 2028; 2030+	F / E	Micron Q1 FY2026 call; TrendForce, 19 Jan 2026
Samsung	P4 and Pyeongtaek expansion	2027–28	E	TrendForce
Industry, 2026	Wafer supply trails demand by >20%, until about 2030		E	SK Group chairman, Mar 2026 (Bloomberg)

2026 wafer shares used by the model (Samsung 32%, SK hynix 27%, Micron 22%, CXMT and others 19%) are our estimates built from trade-press revenue shares: class A.

Exhibit 6: Model assumptions for supply (class A)

Variable	Base	Rationale
Capacity added 2027 / 28 / 29 / 30	Samsung +5/+12/+6/+5%; SK hynix +7/+15/+15/+15%; Micron +5/+11/+8/+6%	Exhibit 5 plans, phased by cleanroom readiness
CXMT and others	+10% a year to 2030, then trend +3 pts fading by 2035	Policy-supported; does not respond to price
Incumbents' response from FY2028 decisions	0.15 / 0.15 / 0.08 × (margin – balance margin); rivals' excess growth offset at 0.5; 2-year lag; -4% to +8% a year	Reproduces the 2018–19 and 2021–23 pattern; not estimated; stress-tested in Exhibit 13

Result (D): industry wafer capacity +6.5% / +12.2% / +9.7% / +9.1% in 2027–30, about 43% cumulatively; Micron's share drifts to about 20%.

2.4 Customer agreements and deposits

Micron has sixteen multi-year agreements covering roughly 20% of DRAM and a third of NAND volume. They carry about \$100bn of cumulative revenue at minimum prices, with ceilings at calendar Q2 2026 prices. We model a floor of at least 60% margin on 22% of DRAM volume through FY2030. Extending the floors to 2045 leaves the requirement unchanged at 69.9%: a 60% floor on a fifth of volume does not bind when the price needs 70% on all of it. Floors protect a glut, not the steady state.

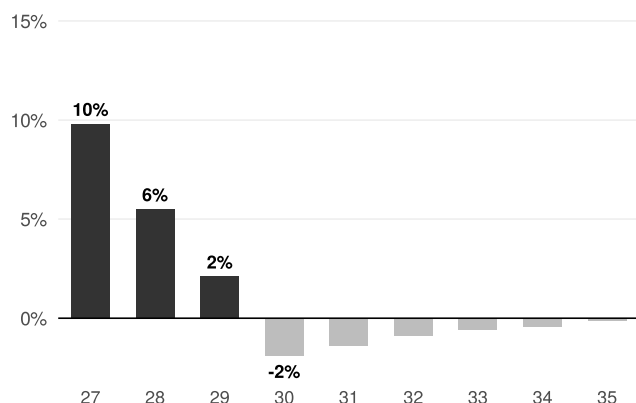
Customers expect to place about \$22bn of deposits and commitments, about \$18bn in cash, returned in the second half of each agreement. Because they are refundable, we treat them as a loan, and value only the interest-free float (about \$6bn, or \$5 a share). If they were never returned, value would rise by at most the present value of the repayments, about \$12bn or \$11

a share (1% of the price).

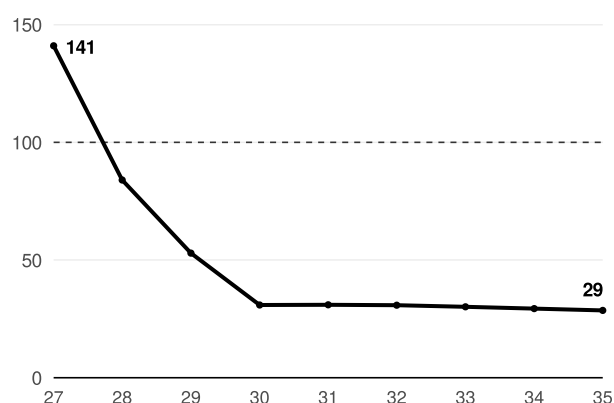
2.5 The base path

Exhibit 7: The gap between wafers needed and wafers installed sets the price per bit; it peaks in 2027 and closes in 2030

DRAM supply gap: wafers required ÷ installed – 1



Conventional DRAM price per bit (FY2026 exit = 100)

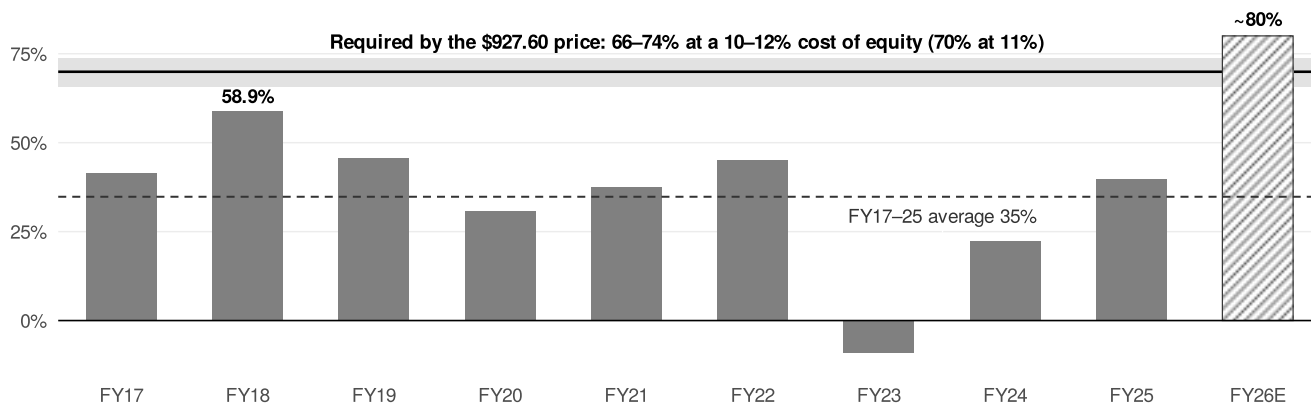


Source: report model, base case at the solved 69.9% margin. Supply gap = wafers required at the balance price ÷ installed wafers – 1. The price premium over the balance price is $(1 + \text{gap})^\beta$ with $\beta = 12$, reached within the year. Class D/A.

FY2027 revenue of \$264bn is 32% above the Q4 FY2026 exit rate: the conventional price per bit averages 41% above the exit level while the gap is at its widest, about 10%. Micron earns about \$192bn of NOPAT (\$168 a share). New fabs land in 2028. By FY2030 announced capacity, growing 9–10% a year, overtakes demand as HBM growth slows to 18%, leaving a surplus of about 2%. The price per bit is then under a quarter of its FY2027 level, and the company margin dips below the steady state before recovering. Working capital, held at 18–20% of revenue, is released as revenue falls, adding \$9–17bn a year in FY2028–30; depreciation trails capex because it catches up over about five years.

2.6 What the price requires

Exhibit 8: After the shortage, the price requires DRAM margins well above the pre-2026 record, permanently



Source: Micron 10-K filings (FY17–25, F); report model. Band: 10–12% cost of equity; line: 11%. FY26E hatched.

Exhibit 9: Less than half the value is cash earned by FY2033

Component (69.9%, 11%)	\$bn	Per share	Share
FCF FY2027–33	438	\$383	41%
FCF FY2034–45	305	\$266	29%
Terminal value	289	\$252	27%
Net cash (24.4) and deposit float (6.1)	31	\$27	3%
Equity value	1,062	\$927.60	100%

Source: report model. Revenue grows about 7.4% a year in FY2033–45 (bit demand ~13% less a falling price per bit); the terminal assumes 3%; at 4% the requirement is 68.9%.

Exhibit 10: Cost of equity and the margin each rate requires

Equity beta	Cost of equity	Required margin
1.10	~10%	65.7%
1.33 (base)	11%	69.9%
1.55	12%	73.6%
1.82 (Micron, Blume-adjusted)	~13%	76.9%

CAPM: 5.00% 10-year Treasury (F) + $\beta \times 4.5\%$ equity risk premium (A). Micron's raw 5-year beta is 2.22 (E). We use a lower beta on purpose: it lowers the requirement, so the conclusion does not depend on a high discount rate.

History supplies a base rate. In the five years after the last record, FY2019–23, Micron's gross margin averaged 30% and never exceeded 45.7%. The price asks for a margin 40 points above that average, held indefinitely.

Exhibit 11: From gross profit to free cash flow at the required margin (\$bn)

Fiscal year	2027	2028	2029	2030	2033	2045
Revenue	264	202	153	105	131	311
Gross margin (company)	88.1%	82.8%	75.7%	62.6%	67.0%	69.8%
Gross profit	233	167	116	65	88	217
Operating expenses	-7	-8	-8	-9	-11	-16
Tax at 15%	-34	-24	-16	-8	-12	-30
NOPAT	192	135	92	48	65	171
Depreciation	14	19	24	25	23	41
Capex	-41	-41	-32	-22	-23	-54
Working capital	-22	17	9	9	-2	-4
Free cash flow	143	130	92	61	63	154

Source: report model. Capex = 13% of revenue + \$13bn per 10% of wafers added. Opex starts at the Q4 FY2026 guide (~\$6.6bn a year) and grows 8% a year to 2033. Tax 15% per Micron guidance. Steady state: 65% operating margin, 64% ROIC; incremental ROIC in FY2040–45 is 70%, consistent with the terminal. Class D/A.

2.7 How long is the shortage? The question the price is really asking

Every serious bull argument reduces to one claim: the shortage is deeper and longer than announced capacity suggests. Exhibit 12 measures that claim directly. Each row changes demand or fab timing, reports when the gap between required and installed wafers closes, and solves again for the margin the price then needs.

Exhibit 12: The longer and deeper the shortage, the lower the bar; a shortage to about 2035 brings the requirement near the record

Scenario	Gap in 2027	Gap closes	Required margin
Base: announced fabs on time; HBM on path	10%	2030	69.9%
Industry fabs one year late	15%	2031	63.7%
Industry fabs two years late	17%	2033	58.0%
HBM demand 20% above path	14%	2034	61.8%
HBM +20% and fabs one year late	20%	2035	53.8%
HBM +40% and fabs one year late	24%	2036	40.9%

Source: report model; 11% cost of equity. For comparison, TrendForce (30 Jul 2026) estimates 2026 DRAM sufficiency at -1% to -2%, widening in 2027; SK Group's chairman (Mar 2026) described a wafer shortfall of more than 20% lasting to about 2030. The model's gap is measured at the balance price, before rationing, so it is larger than a sufficiency ratio.

Taking the chairman at his word, a gap near 20% that runs into the early 2030s, the price needs roughly 54–64%: around the record, not above it. That is the fair version of the bull case, and it is why we do not call the shares overvalued. It needs three things at once: demand ahead of our path, fabs late at several suppliers, and incumbents slow to respond. The last two memory cycles closed their shortages within about two years.

2.8 Can the industry hold 70%? Stress tests on the parameters we set

The 70% is the margin at which value equals price. Whether an industry can hold it is a separate question. At 70% gross margin the business earns a 64% return on capital against an 11% cost of equity, so every supplier has reason to add capacity, including a state-supported entrant.

Exhibit 13: With fabs on schedule, competitive, capital and accounting stresses keep the requirement at 66–76%

Stress (one change at a time; base 69.9%)	Required margin	Note
Competitive behaviour, fabs on announced schedules		
No response to margins at all	68.8%	Announced capacity closes the gap by 2030 on its own
Response $\times 3$; $\times 5$ with a 15% annual cap	70.4%; 72.3%	
CXMT +5% / +15% a year to 2030	66.6% / 72.5%	Export controls that slow CXMT help the bull case
Competitive behaviour if industry fabs slip one year		
Base response; $\times 0.3$; $\times 0.1$	63.7%; 55.2%; 37.4%	A weak response leaves a 3–8% gap in 2035
No response	any	Permanent shortage; no precedent in DRAM
Capital intensity (FY2025 capex: 37% of revenue, 61% of cost of goods; base FY2030–34: 18% and 52%)		
Maintenance capex 25% / 30% of revenue, FY2030–34	71.0% / 71.4%	
Maintenance capex 20% / 25% / 30% of revenue, permanently	72.4 / 74.2 / 76.1%	Steady-state ROIC falls to 48 / 40 / 34%
Accounting and valuation		
Tax 21%; opex +12% a year; shares +5% by dilution	72.2%; 70.5%; 71.4%	
Contract floors extended to 2045	69.9%	Floors below 70% do not bind
Return on new capital 15%; terminal growth 4%	71.3%; 68.9%	
Model parameters		
Price response $\beta = 4$; $\beta = 20$	59.8%; 70.3%	β is a scenario parameter; $\beta = 4$ implies a 2027 gap above 30%
HBM intensity 3.1–3.2 $\times$ (secondary-source HBM4 figure)	66.9%	Not supported by TrendForce's 2027 split

Source: report model (model/outputs_checks.py). Response multiples scale all three incumbents' coefficients in Exhibit 6.

Two points follow. With fabs on schedule, competitive behaviour after 2028 barely matters, because announced capacity closes the gap first; the parameters we set are not what drives the answer. Competition becomes decisive only when fabs slip and incumbents barely respond, which is the bull case of Exhibit 12 restated as behaviour. Heavier reinvestment raises the requirement; it never lowers it.

2.9 What moves the answer

Exhibit 14: Demand and fab timing move the requirement by 10–23 points; everything else by less than 8

Input (class A)	Range	Required margin	Swing (pts)
Conventional DRAM demand growth	10% $\rightarrow$ 16%	80% $\rightarrow$ 56%	23.3
HBM demand ($\times$ path)	0.7 $\rightarrow$ 1.3	77% $\rightarrow$ 56%	20.9
Industry fab delay	0 $\rightarrow$ 2 yrs	70% $\rightarrow$ 58%	11.8
Bits per wafer growth	11% $\rightarrow$ 7%	73% $\rightarrow$ 63%	10.3
Cost of equity	10% $\rightarrow$ 12%	66% $\rightarrow$ 74%	7.9
Cost per bit decline	4% $\rightarrow$ 8%	66% $\rightarrow$ 73%	6.9
NAND demand	13% $\rightarrow$ 22%	71% $\rightarrow$ 67%	3.3

Source: report model. One input at a time. At the 58.9% record margin, the price needs conventional DRAM demand of about 15.5% a year to 2030, or total bit demand of about 18% in 2027 and 16% in 2030; Micron guided to about 20% industry bit growth for calendar 2026.

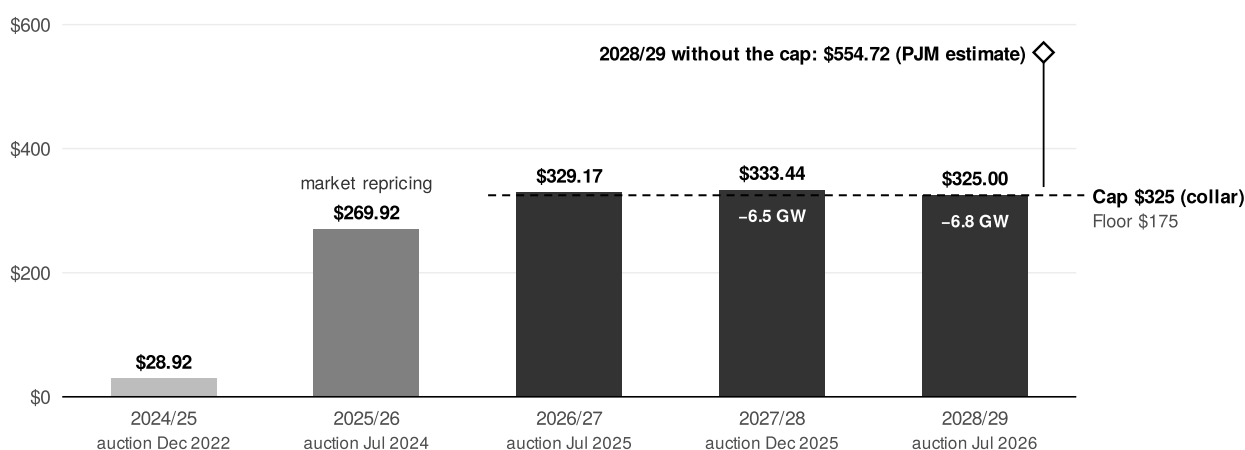
2.10 Bull case and view

Each accelerator generation carries more HBM. Customers are prepaying and signing floor-priced contracts, which is how buyers behave when they expect shortage. New fabs often slip. With conventional demand at 15% and a one-year slip, the price needs about 54%, and the shares look fairly valued. The long-only answer to "six times earnings is cheap" is that 6.0× applies to peak earnings; on the FY2030–33 NOPAT that the price itself requires, \$49 a share, the multiple is 19×. We do not take the bull case as the base, because it needs strong demand, late supply and a weak competitive response together, and none is yet shown by evidence independent of the current shortage. At the current price the modelled margin of safety is limited.

3. Power: a contrast on who captures scarcity

PJM is included as a contrast, not as evidence for the Micron view. It shows the other half of the question: who keeps scarcity value once it is visible. The July 2024 auction repriced capacity from \$28.92 to \$269.92 per MW-day. Every auction since has cleared at the cap of a regulatory collar (\$175–\$325 per MW-day UCAP in 2028/29 terms), which FERC extended through 2029/30 on 28 April 2026. The 2028/29 auction cleared 138,318 MW at exactly the \$325 cap, 6,831 MW short of the reliability requirement; PJM estimates it would have cleared at \$554.72 without the cap, and \$776.69 in the ComEd zone. A Reliability Backstop Procurement opening on 30 September 2026 will contract new resources for up to 15 years at a maximum MW-weighted average of \$555. Existing plants are not eligible. For them the backstop matters by adding supply, which lowers the odds of an uncapped 2030/31.

Exhibit 15: A market repricing in 2024, then clearing at the regulatory cap



Source: PJM Base Residual Auction reports (F); FERC, 195 FERC ¶ 61,076 (28 Apr 2026) (F). RTO prices, \$/MW-day UCAP. The cap was indexed at \$329.17 and \$333.44 for 2026/27 and 2027/28. \$554.72 is PJM's estimate for 2028/29 absent the cap (E).

Exhibit 16: Vistra's merchant exposure to the post-collar price is about 1% of equity per year of outcome

	Pre-Cogentrix	With Cogentrix	Type
Cleared in the 2028/29 auction	10.92 GW	10.92 GW	F
Less Meta 20-year PPAs, 2,176 MW nuclear at 96% ELCC	-2.09 GW	-2.09 GW	F / A
Add Cogentrix PJM plants: 2,048 MW CC at 78%, 1,115 MW CT at 67%		+2.34 GW	F / D
Merchant capacity exposure	8.83 GW	11.17 GW	D
After-tax cash flow per year: at ~\$175 / at ~\$555	-\$0.38bn / +\$0.59bn	-\$0.48bn / +\$0.74bn	D
Equity effect if it lasts one year	-0.6% / +0.9%	-0.7% / +1.1%	D
Equity effect if it lasts five years	-2.4% / +3.6%	-3.0% / +4.6%	D
Equity effect if permanent (upper bound)	-5.6% / +8.7%	-7.1% / +11.0%	D
Probability of ~\$555 needed for +5%, permanent, ~\$175 ruled out	58%	46%	D

Source: Vistra 8-K (14 Jul 2026); Vistra releases (Jan, Aug 2026); PJM 2028/29 ELCC class ratings; POWER Magazine (Cogentrix plant list); model/vistra_exposure.py. Annual after-tax cash flow = GW × 1,000 × (price – \$325) × 365 × (1 – 21%). One year: ÷ 1.1^{3.7}; five years: sum of five such years; permanent: ÷ 10% ÷ 1.1^{3.7}; each ÷ market value of \$47.6bn (336m shares × \$141.53). Probability for +5% = 5% ÷ permanent upside (5 ÷ 8.7 = 58%; 5 ÷ 11.0 = 46%). Tax 21% (A). Retail load (1–3 GW, A) would offset part of the exposure while retail prices are fixed. Cogentrix, FERC-approved on 6 Aug 2026, is expected to close late 2026.

Exhibit 3 answers all seven questions for power; the weak link is question three. Capacity revenue for 2028/29 is already fixed by the auction, and after the collar the regulatory path cannot be identified from public information, so we assign no probabilities. The open value is about 1% of equity for each year an outcome lasts. Isolating it in free cash flow would need unit-level hedge and retail positions that are not public. **No capacity-driven edge**, which is not a view on Vistra's overall valuation. Regulators and contracts, not the physical shortage, decide how much scarcity value reaches merchant equity.

4. Objections and what to watch

These are the objections we expect from people who run the relevant functions: memory procurement at a hyperscaler, accelerator supply chain, foundry advanced packaging, DRAM process engineering, long-only portfolio management, private credit and utility regulatory strategy. They are our formulation, not interviews.

Exhibit 17: Objections, evidence, model treatment and what would change the valuation

Objection (function)	Evidence	Model treatment	What would change the valuation
Agreements extend pricing (memory procurement)	16 agreements; ~\$100bn at minimum prices; ~\$22bn deposits	Floors ≥60% on 22% of DRAM; extended to 2045, no change (69.9%)	Renewals with floors near 70% on most of the volume
HBM per accelerator is rising faster (accelerator supply chain)	HBM 22% → 30% of wafer input, end-2026 to end-2027; HBM4 12-high ramp twice as fast as HBM3E	HBM +20%: gap closes 2034, requirement 61.8%	HBM above 35% of wafer starts in 2028 with fabs slipping
Packaging, not wafers, limits HBM (advanced packaging)	Interposer and base-die logic capacity	Acts as HBM demand below 1×: at 0.8×, 74.6%	Packaging caps HBM through 2028: bearish for conventional DRAM, not HBM prices
Bit growth per wafer is slowing (DRAM process)	EUV-heavy nodes; 12–16-high stacks	9% a year; at 7%, 62.8%	Node transitions slipping at two or more suppliers
Six times forward earnings is cheap (long-only)	$\$927.60 \div \$155 = 6.0\times$	19× on the FY2030–33 NOPAT the price requires (\$49/share)	Evidence of post-cycle margins above 60%
Deposits are not debt (private credit)	~\$18bn cash, refundable in the second half of each term	Loan; float \$5/share; if never returned, +\$11/share	Forfeitures, or conversion into non-refundable prepayments
The backstop caps post-2030 prices (utility regulatory)	Up to 15-year contracts; \$555 maximum average	No probabilities; ~1% of equity per year	Little new supply contracted, and a 2030/31 design without a collar

Exhibit 18: What to watch: dated events, leading indicators and the post-2030 test

When	Indicator	Against our view	For our view
30 Sep 2026	Micron FY2026 results, FY2027 outlook	FY2027 EPS well above ~\$168, prices re-accelerating	Guide in line; pricing commentary flattening
Quarterly	Hyperscaler capex guidance; accelerator vendors' data-centre revenue and HBM per package	Capex still rising into 2028	Capex growth slowing in 2027
Quarterly	TrendForce DRAM sufficiency; spot versus contract prices	Sufficiency worsening beyond –5%	Spot below contract in 1H CY2027
1H CY2027	Yongin tool-in; ID1 first wafer; Tongluo ramp; HBM4 qualification at the leading accelerator vendor	Milestones a year late at two or more suppliers	Milestones on time
2027	PJM backstop results; 2030/31 market design	No collar and a persistent shortfall	New supply contracted; collar-like design
FY2029–31	Contract renewal terms; Micron capex per wafer; FY2030 gross margin	Renewals near 70%; margin above 65% in a surplus year	FY2030 margin below 60%

All data as of 15 Sep 2026. Limits: nothing here is out of sample. The model treats DRAM as one market, prices HBM at wafer parity, and does not model customer inventory swings or a disorderly supplier exit.

Appendix A: Model inputs

Input	Base	Range tested	Basis	Type
Starting point	\$200bn revenue at 85% GM; DRAM 76%; opex ~\$6.6bn; 1.145bn shares		Q4 FY2026 guidance annualised; Q3 mix	F / D
Conventional DRAM bit demand	13% a year	9–17%	Micron: ~20% industry bit growth in CY2026 incl. HBM (F)	A
HBM bit demand	45%, 30%, 22%, 18% (2027–30), then to 13%	× 0.6–1.4	Micron: HBM TAM ~40% CAGR to 2028 (F)	A
HBM wafer per bit vs rest of DRAM	2.85 (2026); 2.87 (2027–29); 2.85 (2030); 2.8 from 2031	up to 3.2	TrendForce wafer and bit shares, end-2026 and end-2027	D
HBM cost overhead	35%	20–50%	Stack yield, TSV, base die	A
Supply schedules and reactions	Exhibits 5–6	Exhibit 13	Announced plans; judgement	E / A
Bits per wafer; cost per bit	9% (8% after 2033); –6% (–5% after 2033)	7–12%; –4 to –8%	Node migration	A
Price response	$\beta = 12$; $\lambda = 1$	β 4–20; λ 0.6–1	Scenario parameter	A
NAND	Demand 17%; bits/wafer 18%; balance margin 10 pts below DRAM	demand 13–22%	TrendForce: NAND sufficiency positive in 2027	E / A
Contracts	≥60% on 22% of DRAM to FY2030; ceiling at FY2026 exit	to 2045	Micron Q3 FY2026 remarks	F / A
Capex; working capital	13% of revenue + \$13bn per 10% wafers; 20% → 18% of revenue	maintenance 10–30%	FY2025–26 capex	A
Tax; opex growth	15%; +8% a year to 2033, then 3%	21%; +12%	Micron Q4 FY2026 guidance (tax)	F / A
Return on new capital	Steady-state ROIC, 64% (model and Exhibit 2)	15–30%	Incremental ROIC FY2040–45: 70%	D
Terminal	FY2045; $g = 3\%$	2040–50; g 4%		A
Cost of equity	11% (β 1.33)	10–13%	5.00% UST + $\beta \times 4.5\%$ ERP	F / A
Net cash; deposits	\$24.4bn; \$18bn refundable, repaid FY2029–31	never repaid	28 May 2026 balance sheet; Q3 remarks	F

Every number is written by model/outputs.py, outputs_checks.py and vistra_exposure.py to data/numbers.json, the single source for both language editions. Each input has a record in data/fact_ledger.json with source, locator, tier, class, cross-check and status.

Appendix B: Sources

Data	Primary source	Cross-check	Tier
Q3 FY2026 revenue \$41.5bn; DRAM \$31.3bn, prices +low 60s%, bits +low single digits; Q4 guide \$50.0bn, ~86% GM, ~15% tax, ~1.15bn shares; 16 agreements, ~\$100bn, \$22bn / \$18bn deposits; FY2026 capex ~\$27bn	Micron Q3 FY2026 release and prepared remarks, 24 Jun 2026	10-Q (Q3 FY2026); Investing.com slide summary	T1
FY2018 revenue \$30.39bn, GM 58.9%, Q4 GM 61.0%, operating income \$14.99bn; FY2025 revenue \$37.38bn, operating income \$9.77bn, capex \$13.80bn; annual GAAP gross margins FY2017–25 (41.5, 58.9, 45.7, 30.6, 37.6, 45.2, –9.1, 22.4, 39.8%)	Micron fourth-quarter results releases, FY2017–25 (Exhibit 99.1 filings)	10-K filings	T1
HBM 22% / 30% of wafer input, 9% / 13% of bits (end-2026 / end-2027)	TrendForce, "Tight DRAM Supply Gives Suppliers Greater Pricing Power in HBM...", 2 Jun 2026	Industry coverage	T2
DRAM sufficiency –1% to –2% (2026), widening in 2027; NAND positive in 2027	TrendForce, "Diverging Memory Market Outlook in 2027...", 30 Jul 2026	Industry coverage	T2
Tongluo phase 1 >10% of Micron capacity from 2H 2027	TrendForce, "Micron's Acquisition of PSMC's Tongluo Fab...", 19 Jan 2026	Micron Q1 FY2026 call	T2
Wafer supply trails demand by >20% until about 2030	Chey Tae-won, SK Group, NVIDIA GTC, Mar 2026	Taipei Times; Tom's Hardware	T2

Data	Primary source	Cross-check	Tier
	(Bloomberg, 17 Mar 2026)		
BRA prices; 2028/29: 138,317.8 MW, 6,831.3 MW short, \$554.72 RTO and \$776.69 ComEd uncapped; cap \$325, floor \$175	PJM 2028/2029 BRA Report, 14 Jul 2026	PJM news release	T1
Collar extended to 2028/29 and 2029/30	FERC, 195 FERC ¶ 61,076, 28 Apr 2026	FERC, 195 FERC ¶ 61,211	T1
Backstop: up to 15 years, \$555 maximum weighted average, 6,831 MW less bilateral contracts, opens 30 Sep 2026	PJM Board decision; PJM Inside Lines	American Public Power Association	T1
Vistra 10,924 MW cleared 2028/29 at \$325; 336m shares; hedges 100 / 94 / 72%	Vistra 8-K, 14 Jul 2026; Q2 2026 release	StockTitan filing summary	T1
ELCC ratings: nuclear 96%, gas CC 78%, gas CT 67%	PJM, ELCC Class Ratings for the 2028/2029 BRA		T1
Meta PPAs 2,176 MW + 433 MW uprates, 20 years; Cogentrix 5.5 GW, PJM plant list, FERC approval 6 Aug 2026	Vistra releases, 5 and 9 Jan, 7 Aug 2026	POWER Magazine; RTO Insider	T1/T2
MU \$927.60, VST \$141.53 (15 Sep 2026); UST 10y 5.00%	Exchange closing prices; FRED DGS10	Two price histories	T1
FY2027 consensus EPS \$155 (7 Sep 2026); MU 5-year beta 2.22	Secondary data providers	Replace with Bloomberg values as of 15 Sep 2026 before distribution	T3

T1: filings, market operators, regulators. T2: industry sources with explicit figures. T3: secondary coverage, used to locate evidence but not to settle a question.

Disclosures

Nature of this report. For research and educational purposes only. Not investment advice, an offer, or a recommendation to buy or sell any security. No rating and no price target.

View labels. "No demonstrated valuation edge" means the operating state implied by the price is not established by the public evidence reviewed, and the analysis does not identify a return above the cost of equity; it is not a sell view. "No capacity-driven edge" means the unpriced part of the capacity question is small relative to equity value or cannot be isolated from public data. Neither is a rating.

Data and models. Market data as of the 15 September 2026 close. Later events are noted where material and are not model inputs. Forward-looking figures are model outputs under stated assumptions.

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